

Day 4: Bren Math Workshop

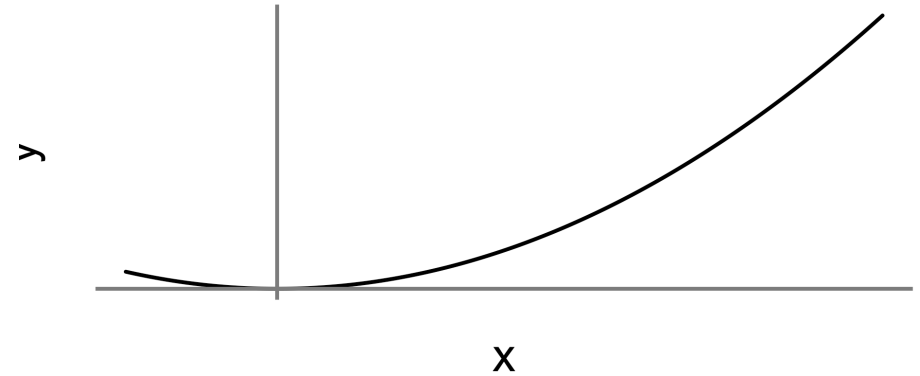
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Interpreting the slope

Slope (average)

Sometimes, it can be useful to find the **average rate of change** of a function.



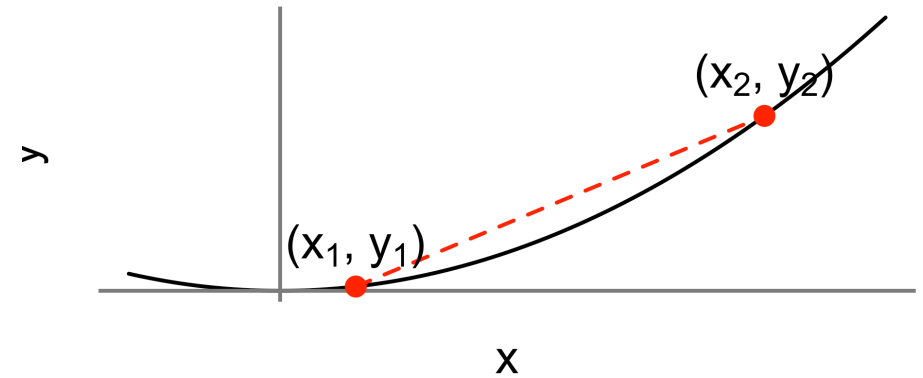
Slope (average)

Sometimes, it can be useful to find the **average rate of change** of a function.

Between any two points (x_1, y_1) and (x_2, y_2) on the graph of a function, the slope is found by:

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Same idea as in a line!



Slope represents rate of change

Ice-core data before 1958. Mauna Loa data after 1958.

Source: [The Keeling Curve](#)

Get into the practice of saying the meaning out loud

- As if you're explaining it to someone unfamiliar with the data
- Including units
- Without overstating certainty

For example:

“Between 1972 and 2020 the price of hobbit homes increased by an average of \$2,450 per year”

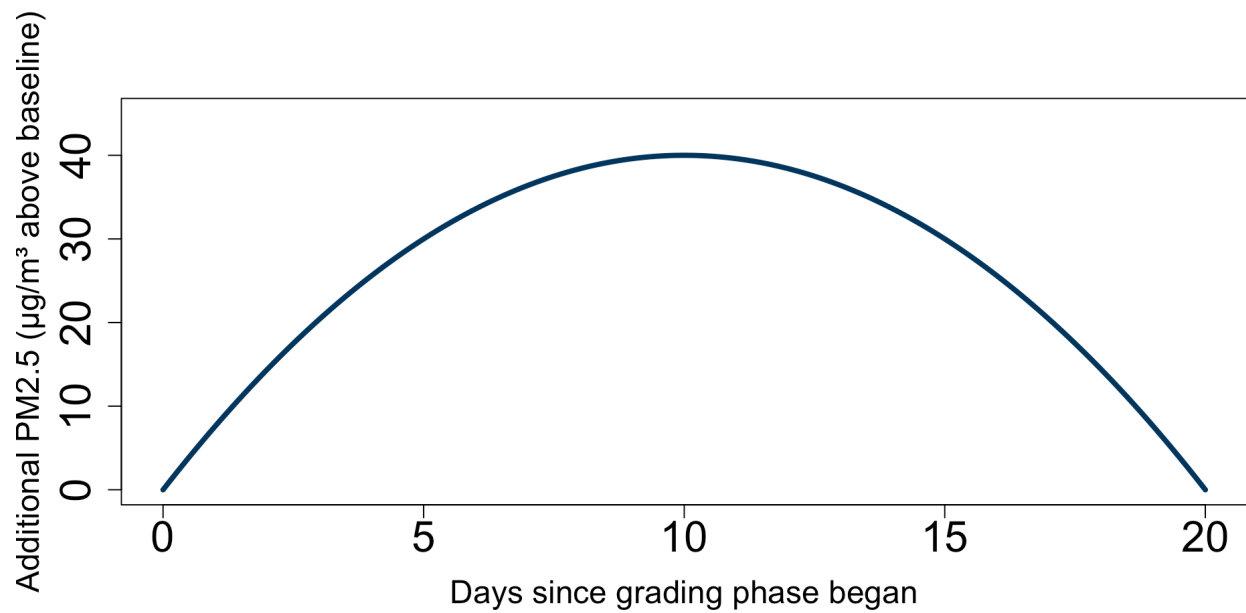
differs from

“Between 1972 and 2020 the price of hobbit homes increased by \$2,450 per year.”

Slope practice

 A community air-quality monitoring group tracked the additional fine particulate matter concentration, $P(t)$ ($\mu\text{g}/\text{m}^3$ of PM2.5 above baseline), in the neighborhood's air over 20 days of a construction phase.

t (days)	$P(t)$ ($\mu\text{g}/\text{m}^3$)
2	14.4
6	33.6
10	40.0
14	33.6
18	14.4

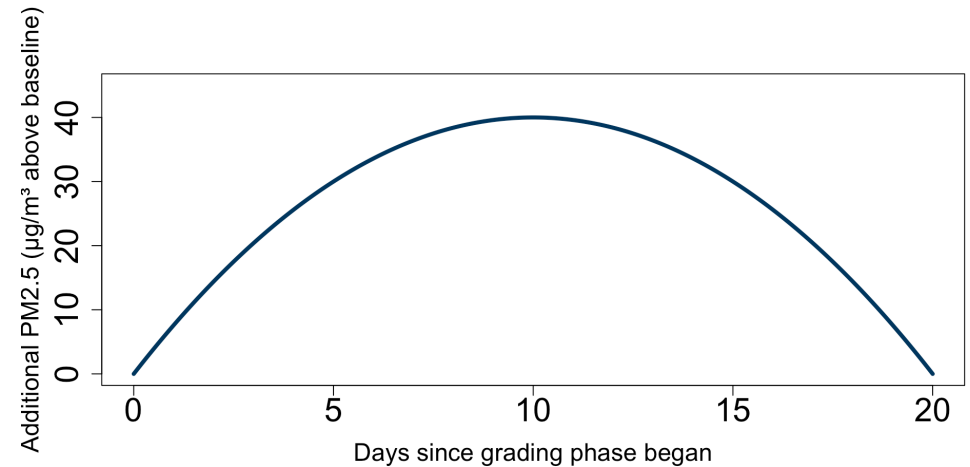


- a. Calculate the average rate of change in additional PM2.5 concentration between day 2 and day 6. Represent this graphically on the plot.
- b. Write your answer as a full sentence, with units, without overstating certainty.

Slope practice

$P(t)$ measures $\mu\text{g}/\text{m}^3$ of PM2.5 above baseline in the neighborhood's air over the 20 days of that construction phase.

t (days)	$P(t)$ ($\mu\text{g}/\text{m}^3$)
2	14.4
6	33.6
10	40.0
14	33.6
18	14.4



- a. Calculate the average rate of change in additional PM2.5 concentration between day 2 and day 6. Represent this graphically on the plot.
- b. Write your answer as a full sentence, with units, without overstating certainty.



Slope practice

a. average rate of change = $\frac{P(6) - P(2)}{6 - 2} = \frac{33.6 - 14.4}{4} = \frac{19.2}{4} = 4.8 \frac{\mu\text{g}/\text{m}^3}{\text{day}}$

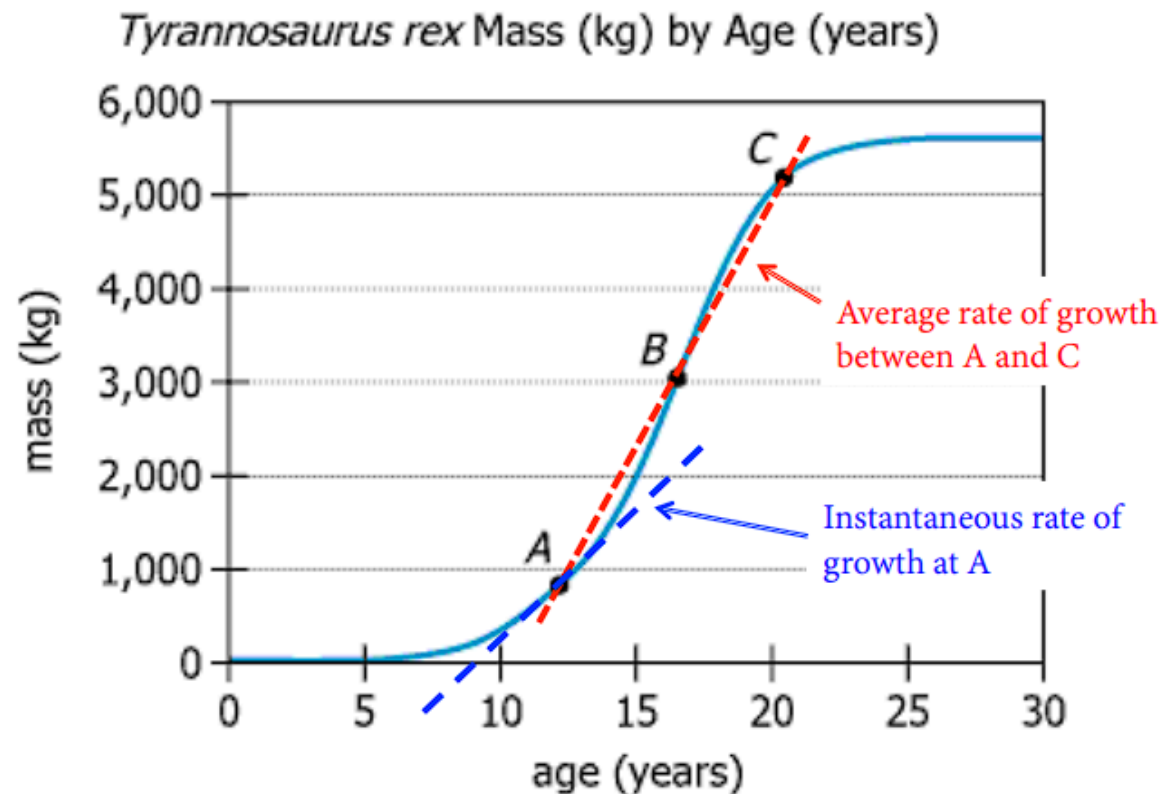
b. Between day 2 and day 6 of the grading phase, additional PM_{2.5} concentration in the neighborhood increased by an average of 4.8 $\mu\text{g}/\text{m}^3$ per day.

The *average* slope of a continuous function rarely tells the whole story



Slope can be *average* or *instantaneous*

- Rise over run can always be used to find average rate of change between two parts of a graph
- Instantaneous rate of change leads us to ✨calculus✨.



Limits

Limits



Limits

Definition

We say a number L is the limit of a function $f(x)$ as its variable x approaches a number c , if the function's output values $f(x)$ approach L when x get closer to c .

We write this symbolically as:

$$\lim_{x \rightarrow c} f(x) = L$$

We read this as: “The limit of $f(x)$ as x approaches c is L .”

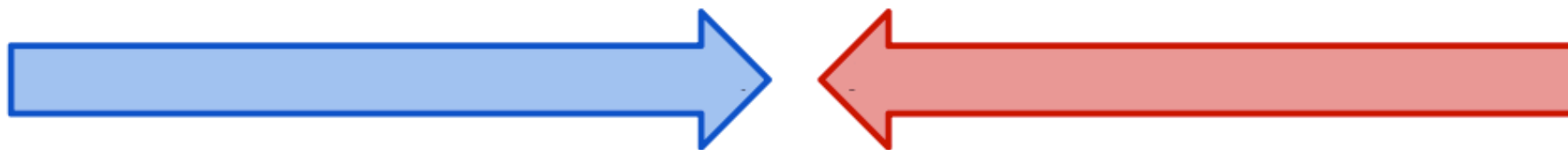
Limits example: numerical

What is the limit of $f(x) = 2x^2 - 4$ as x approaches 2?

We need to examine the values of $f(x)$ as x gets closer to 2 from both sides.



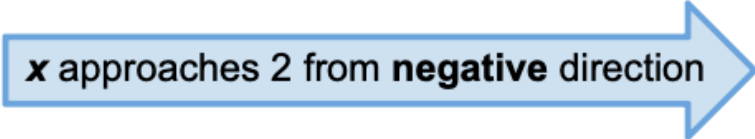
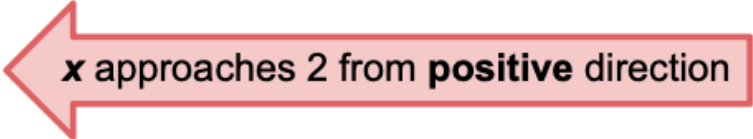
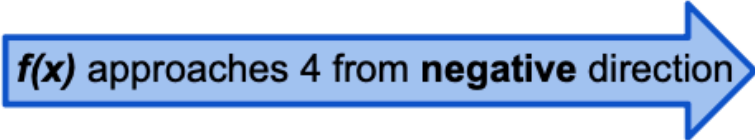
x					2				
$f(x)$									



Limits example: numerical

What is the limit of $f(x) = 2x^2 - 4$ as x approaches 2?

We need to examine the values of $f(x)$ as x gets closer to 2 from both sides.

 x approaches 2 from negative direction					 x approaches 2 from positive direction				
x	1.9	1.99	1.999	1.9999	2	2.0001	2.001	2.01	2.1
$f(x)$	3.22	3.9	3.99	3.999	4	4.0008	4.008	4.08	4.8
 $f(x)$ approaches 4 from negative direction					 $f(x)$ approaches 4 from positive direction				

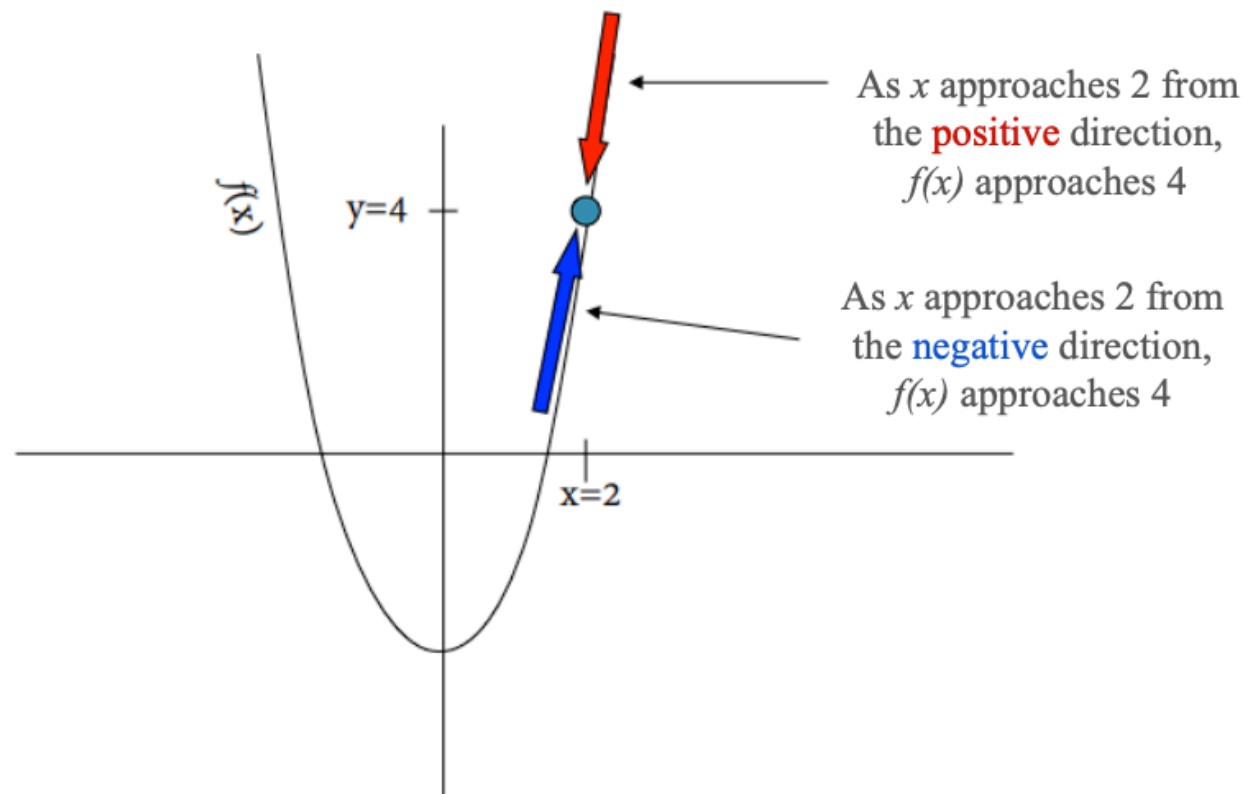
From both directions it looks like $f(x)$ converges to 4.

So, we say that “The limit of $2x^2 - 4$ as x approaches 2 is 4”. We can write it like this:

$$\lim_{x \rightarrow 2} (2x^2 - 4) = 4$$

Limits example: graph

What is the limit of $f(x) = 2x^2 - 4$ as x approaches 2?

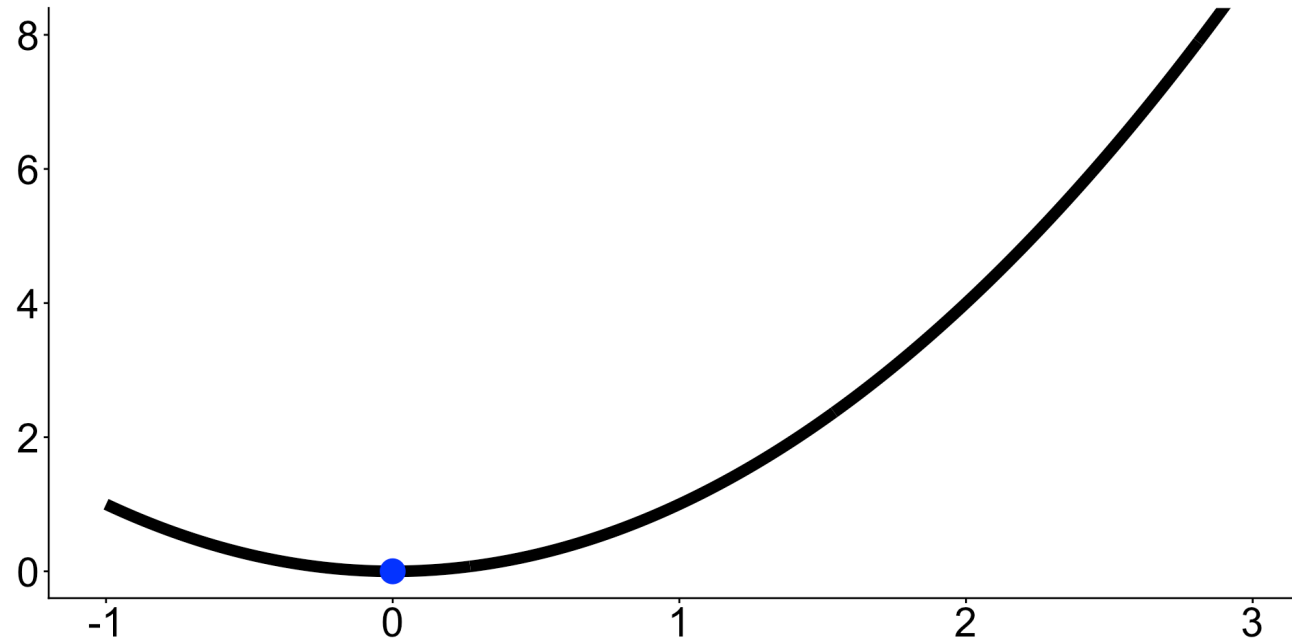


From both directions it looks like $f(x)$ converges to 4.

So, $\lim_{x \rightarrow 2}(2x^2 - 4) = 4$.

Introduction to derivatives

Average slope

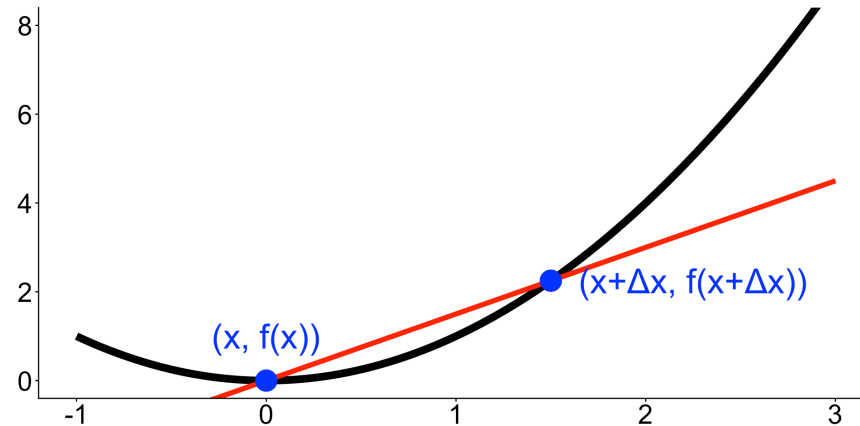


Start with:

Nearby point:

Average slope:

Average slope



Start with a point $(x, f(x))$ on the graph.

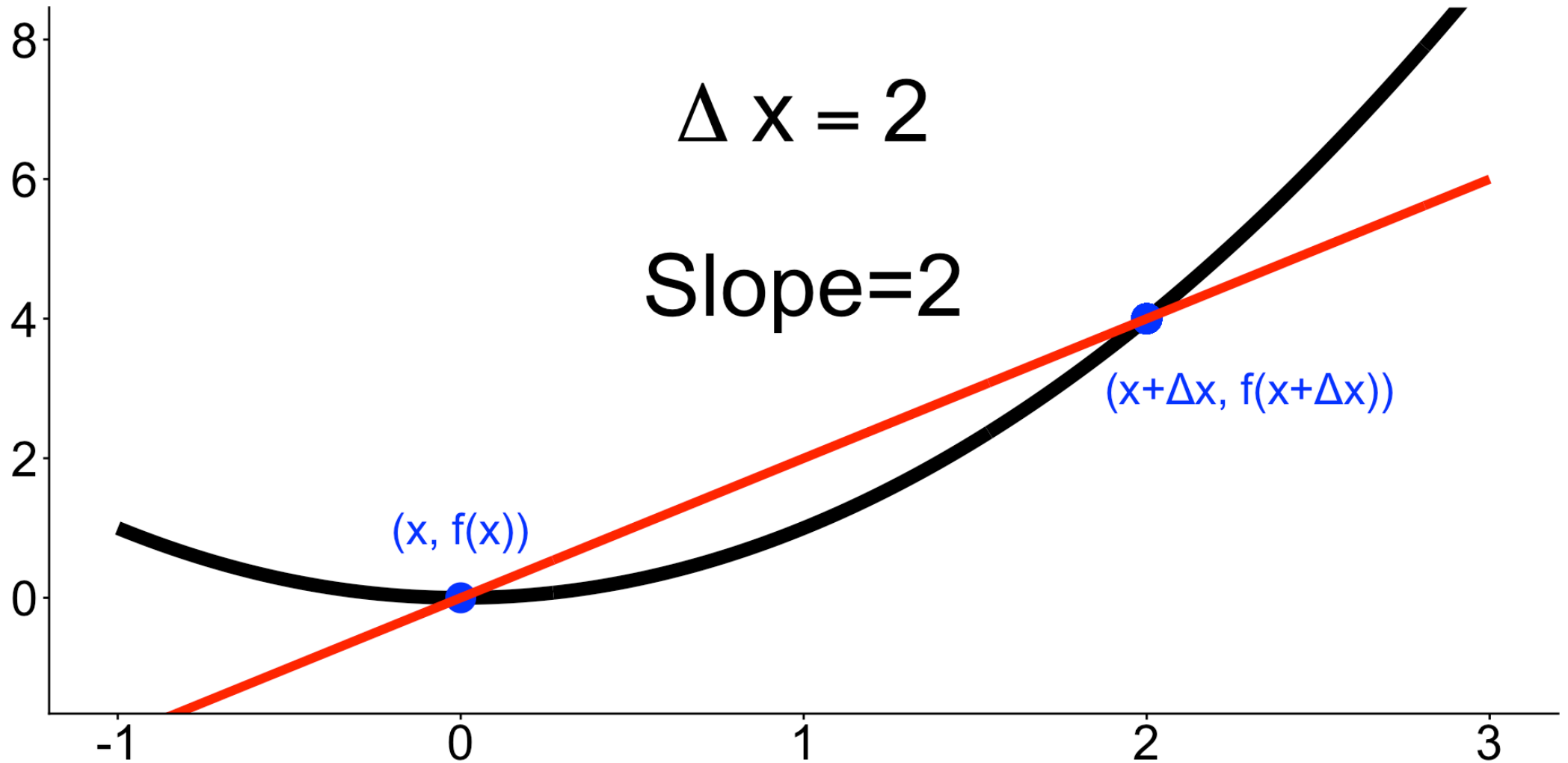
A nearby point can be $(x + \Delta x, f(x + \Delta x))$, where Δx is a short distance from x .

The average slope between $(x, f(x))$ and $(x + \Delta x, f(x + \Delta x))$ is given by

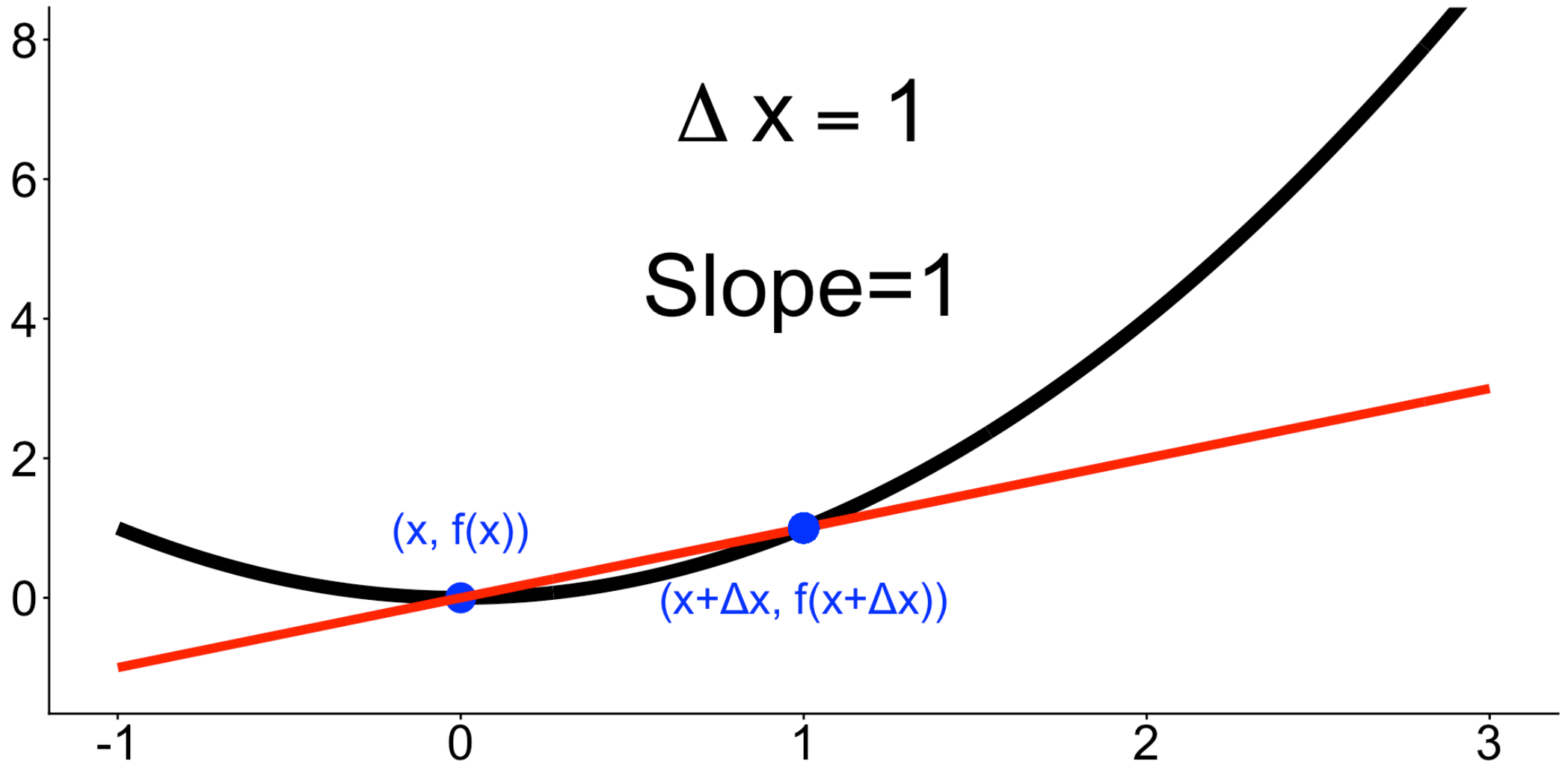
$$\text{average slope} = \frac{f(x + \Delta x) - f(x)}{(x + \Delta x) - x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

What happens when Δx becomes smaller and smaller?

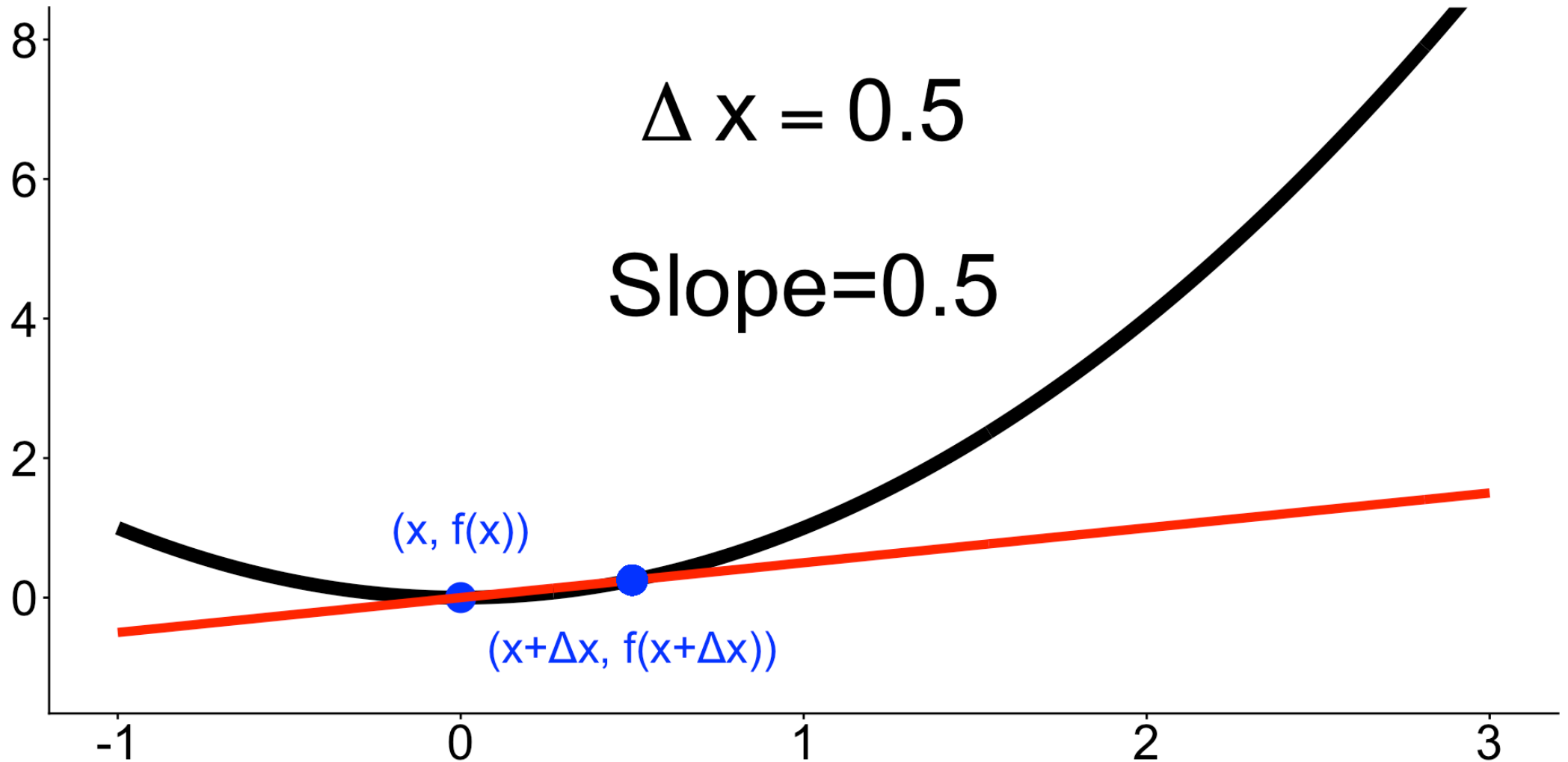
What happens when $\Delta x \rightarrow 0$?



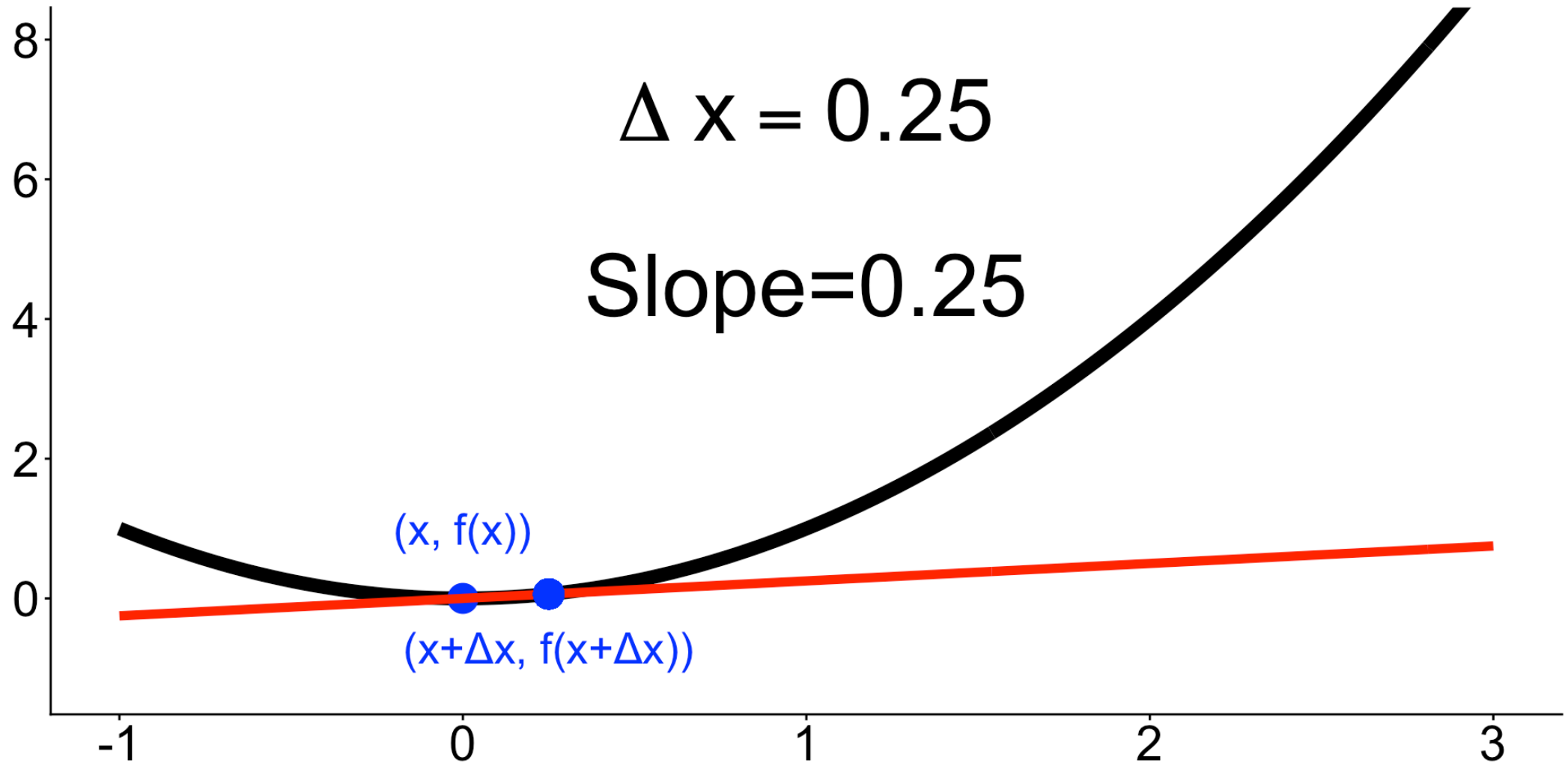
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What happens when $\Delta x \rightarrow 0$?



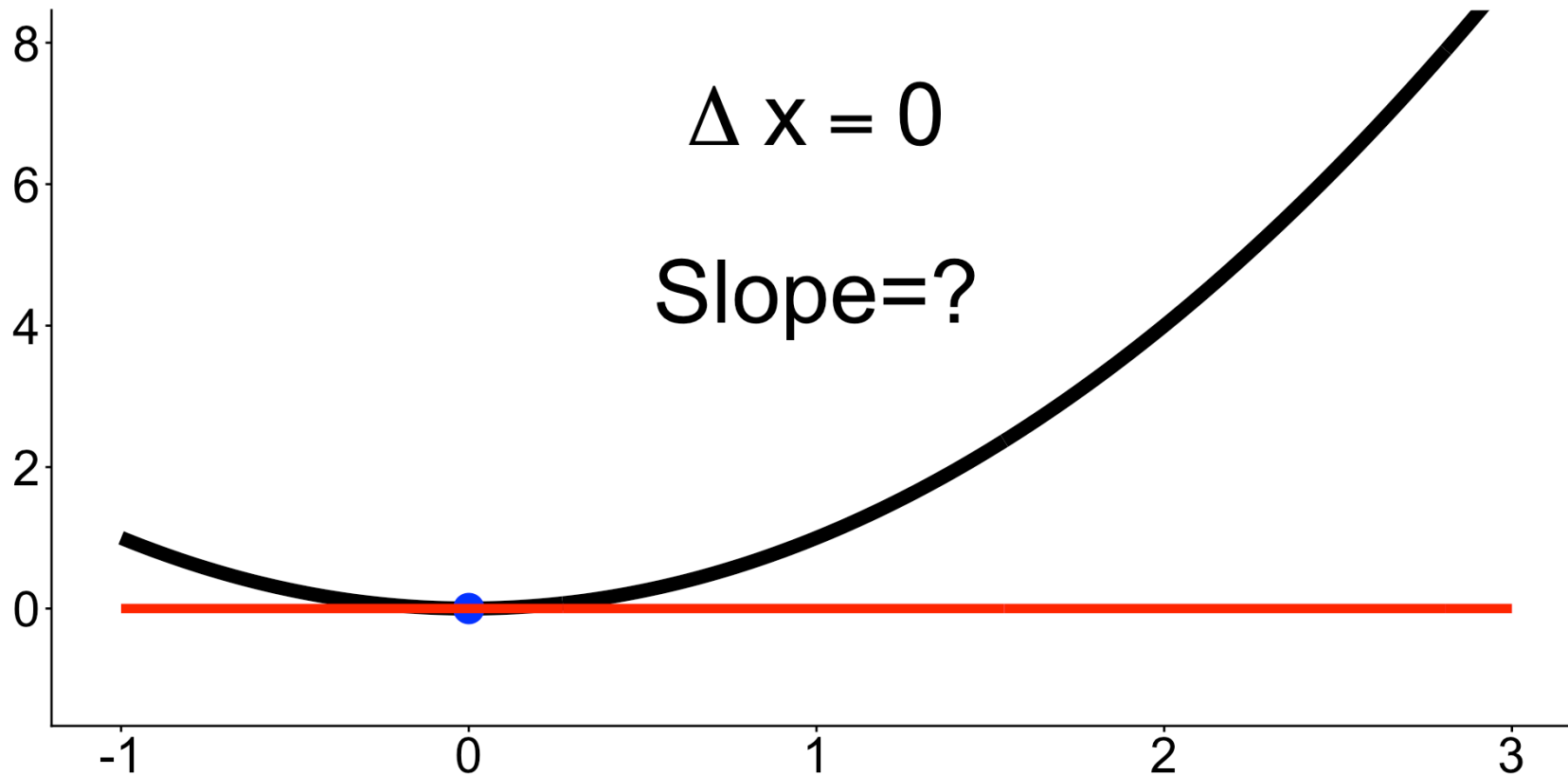
What happens when $\Delta x \rightarrow 0$?



Tangent lines

What if we let $\Delta x = 0$?

Then we would have the slope of a line that only touches the graph at exactly $(x, f(x))$.



These very special types of lines are called **tangent lines**.

Derivative definition

Remember the slope of the line that passes between two points (x_1, y_1) and (x_2, y_2) is:

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}.$$

Derivative definition

Remember the slope of the line that passes between two points (x_1, y_1) and (x_2, y_2) is:

$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1}.$$

If we have a function $f(x)$, then $(x, f(x))$ represents a point on its graph.

Choose a different point on the graph that is Δx away: $(x + \Delta x, f(x + \Delta x))$.

Derivative definition

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Choose a different point on the graph that is Δx away: $(x + \Delta x, f(x + \Delta x))$.

Plug these points on the graph of $f(x)$ into the slope equation:

$$\text{slope} = \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

Derivative definition

Remember the slope of the line that passes between two points (x_1, y_1) and (x_2, y_2) is:

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If we have a function $f(x)$, then $(x, f(x))$ represents a point on its graph.

Choose a different point on the graph that is Δx away: $(x + \Delta x, f(x + \Delta x))$.

Plug these points on the graph of $f(x)$ into the slope equation:

$$\text{slope} = \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

When we let $\Delta x \rightarrow 0$ we get **the slope of the tangent line at x**

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

This is the derivative of f at x . Also denoted $\frac{df}{dx}$.

The expression for the instantaneous slope at any point on a function, aka the **derivative**

IS FOUND BY:

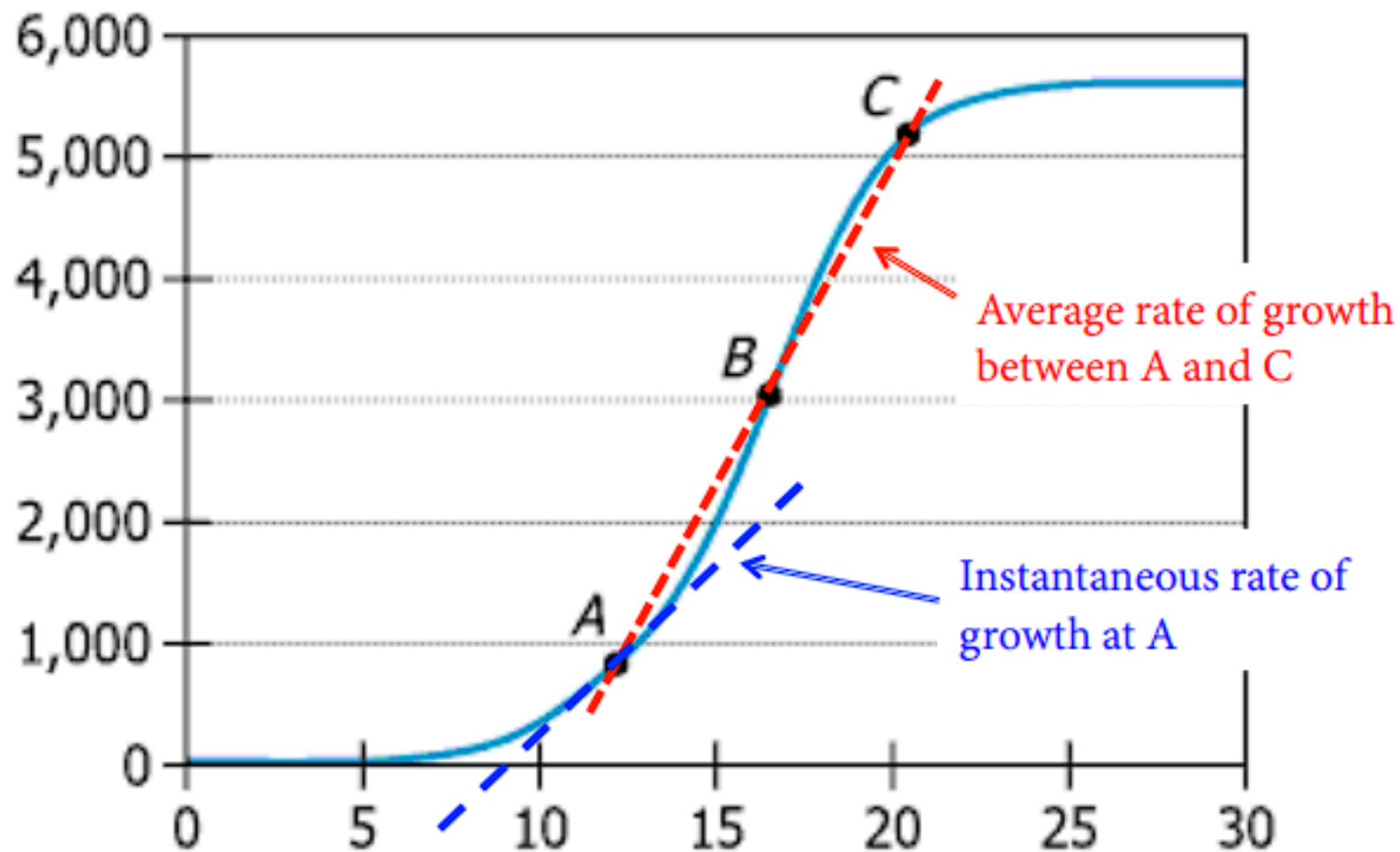
- ① Finding an expression for the **slope** between 2 points separated by Δx ...

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

- ② evaluating that slope as the points get infinitely close together.



Calculus and derivatives are used to study change



Environmental Science studies change



🤔 Talk with a partner, what are 2 fields of environmental science where studying the rate of change and derivatives would be important?

Let's take a 10 minute break

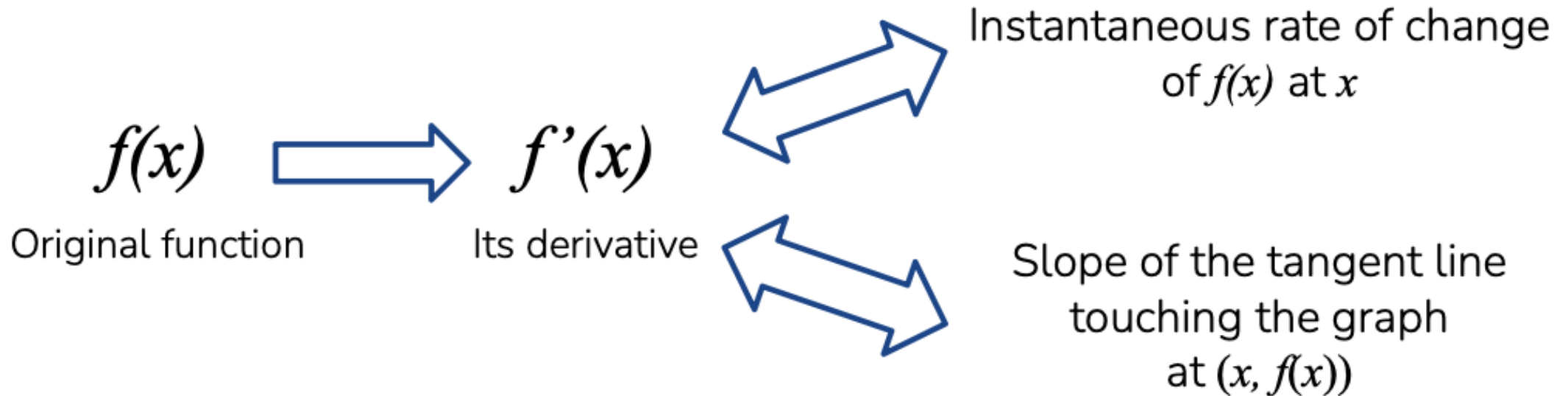


image: Flaticon.com

Interpreting the derivative

Relationship between functions and their derivatives

We interpret it as:



Remember: $f'(x) = \frac{df}{dx}$ is the derivative of $f(x)$. Same thing, different notation!

Units of the derivative

If we have a function $f(x)$, then

$f'(x)$ is approximately equal to the change in $f(x)$ when the variable x increases by 1 unit.

Example:

- $g(t)$ = temperature in degrees Celsius at t hours after midnight
- $g(7) = 15$ means
- $g'(7) = 3$ means
- $g'(7) = -2$ means

Units of the derivative

If we have a function $f(x)$, then

the units for $f'(x)$ are the units of $f(x)$ divided by the units of x .

$f'(x)$ is approximately equal to the change in $f(x)$ when the variable x increases by 1 unit.

Example:

- $g(t)$ = temperature in degrees Celsius at t hours after midnight
- $g(7) = 15$ means the temperature is 15°C at 7 hours after midnight (7 am)
- $g'(7) = 3$ means the temperature is increasing at a rate of $3^{\circ}\text{C}/\text{hour}$ at 7am
- $g'(7) = -2$ means the temperature is decreasing at a rate of $2^{\circ}\text{C}/\text{hour}$ at 7am

Units of the derivative



The cost of extracting T tons of ore from a copper mine is $C = f(T)$ dollars. What does it mean to say that $f'(2000) = 100$?

Write your answer as a full sentence, with units.

Units of the derivative



The cost of extracting T tons of ore from a copper mine is $C = f(T)$ dollars. What does it mean to say that $f'(2000) = 100$?

Write your answer as a full sentence, with units.

- Units of f' are dollars per ton, so $f'(2000) = 100$ dollars per ton.
- When 2000 tons have been extracted, the cost of extraction increases at a rate of \$100 per ton.
- When 2000 tons of ore have already been extracted from the mine, the cost of extracting the next ton is approximately \$100 more.

Sign of the derivative

- If the **derivative is positive**, it means the rate of change is positive, and the function is creasing.
- If the **derivative is negative**, it means the rate of change is negative, and the function is decreasing.
- If the **derivative is zero**, it means the rate of change is zero, and the function is neither increasing nor decreasing (it **may be at a local maximum or minimum**).

Sign of the derivative

- If the **derivative is positive**, it means the rate of change is positive, and the function is increasing.

$$f'(a) > 0 \implies \text{function is increasing near } a$$

- If the **derivative is negative**, it means the rate of change is negative, and the function is decreasing.

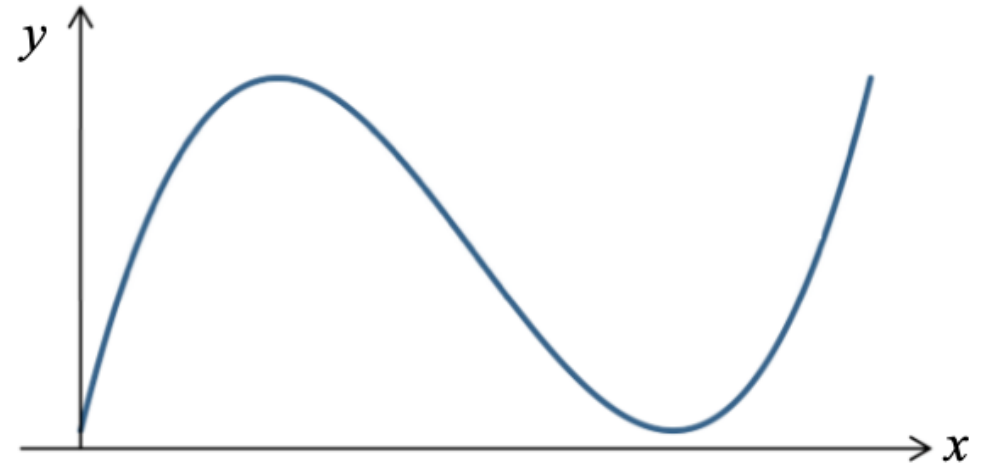
$$f'(a) < 0 \implies \text{function is decreasing near } a$$

- If the **derivative is zero**, it means the rate of change is zero, and the function is neither increasing nor decreasing (it **may be at a local maximum or minimum**).

Sign of the derivative practice



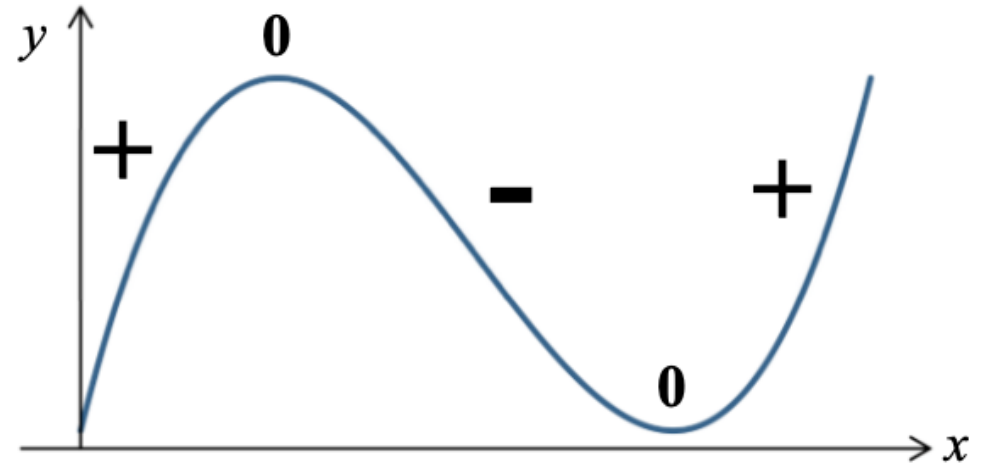
1. Determine for which sections of the graph of $f(x)$ the derivative is positive, negative, or zero.



Sign of the derivative practice



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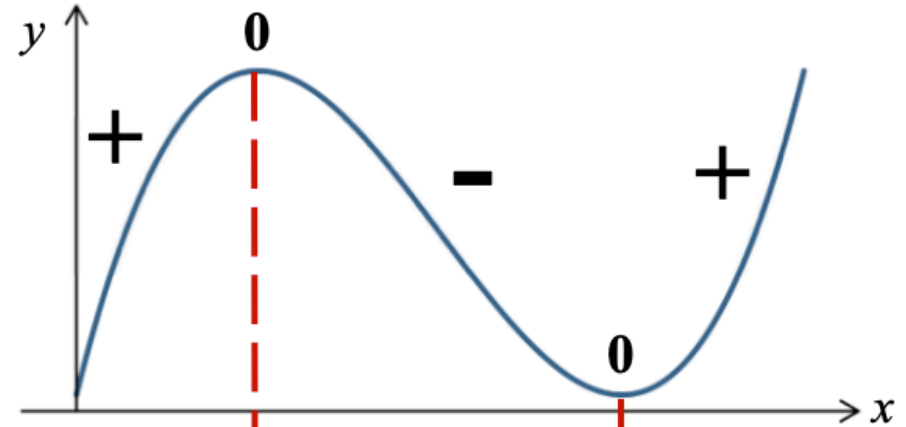


2. *Example:* Sketch a graph for the derivative.

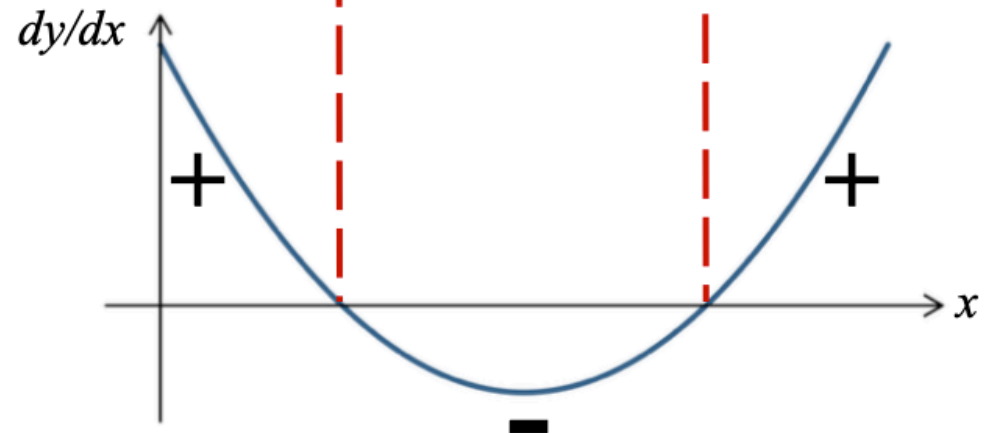
Sign of the derivative practice



1. Determine for which sections of the graph of $f(x)$ the derivative is positive, negative, or zero.



2. *Example:* Sketch a graph for the derivative.

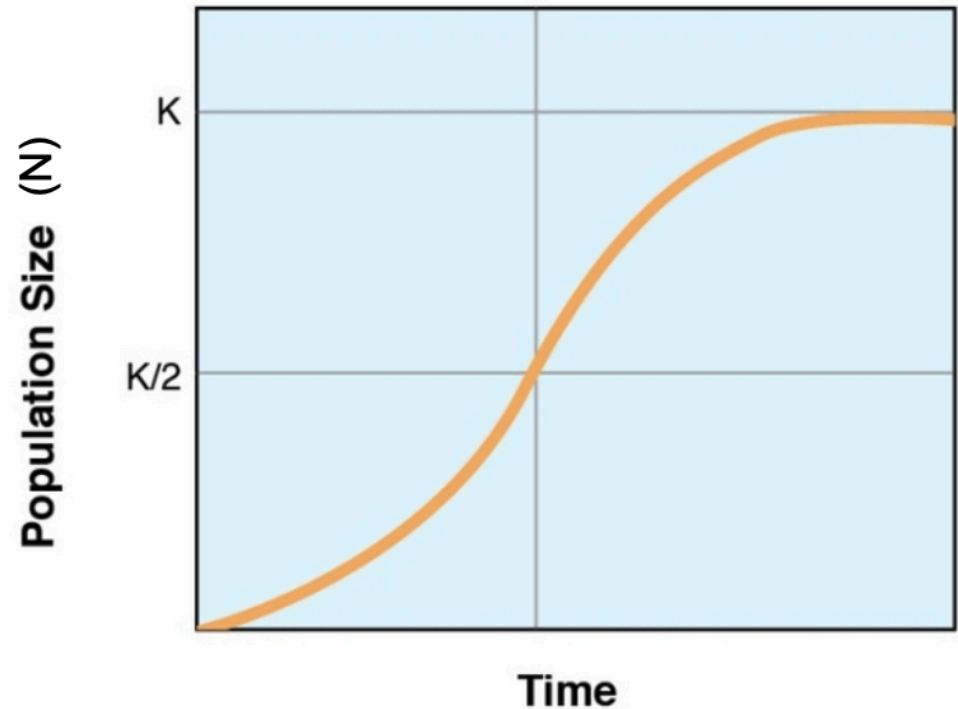


Application: Logistic growth

In logistic growth, the population N grows rapidly at first, but then slows down as it approaches a maximum population size K , called the carrying capacity.

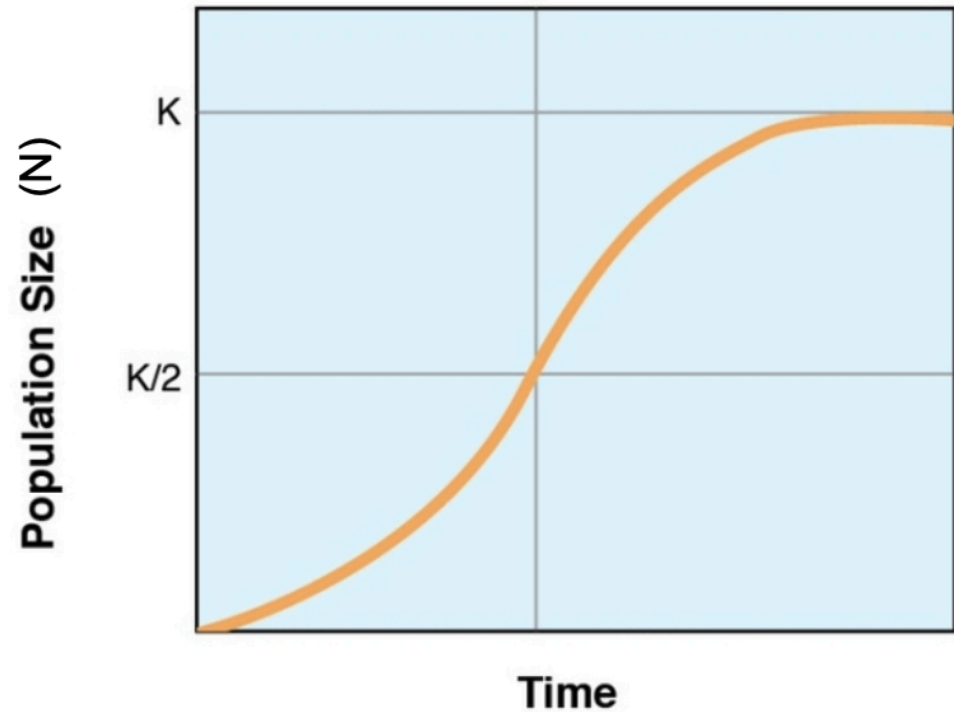


1. What does it mean that $\frac{dN}{dt}$ is positive, negative, or zero in terms of the population growth?
2. When do you think $\frac{dN}{dt}$ is positive, negative, or zero?
3. When is population growth fastest?



Application: Logistic growth

1. $\frac{dN}{dt} > 0$ means the population is growing; $\frac{dN}{dt} < 0$ means the population is shrinking; $\frac{dN}{dt} = 0$ means the population size is momentarily unchanged.
2. $\frac{dN}{dt}$ is **positive** for all $0 < N < K$, since the population is always increasing toward the carrying capacity. $\frac{dN}{dt}$ is **zero** at $N = 0$ and $N = K$, the two points where the population stops changing.
3. Population growth is **fastest** at $N = K/2$, the inflection point of the curve, where the slope of $N(t)$ is steepest.



Rules for differentiation

Rules for differentiation (1)

Constant Rule 

Power Rule 

Rules for differentiation (1)

Constant Rule

Let c be a constant (so, a number). If $f(x) = c$, then $f'(x) = 0$. Equivalently

$$\frac{d}{dx}c = 0$$

Power Rule

Let n be a (real) number, if $f(x) = x^n$, then $f'(x) = nx^{n-1}$. Equivalently

$$\frac{d}{dx}[x^n] = nx^{n-1}.$$

Examples

$$y = 100 \qquad f(x) = x^5 \qquad f(x) = \frac{1}{x^2}$$

Rules for differentiation (2)

Sum and Difference Rules



Rules for differentiation (2)

Sum and Difference Rules

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\begin{aligned}\frac{d}{dx}[f(x) + g(x)] &= \frac{d}{dx}f(x) + \frac{d}{dx}g(x) \\ \frac{d}{dx}[f(x) - g(x)] &= \frac{d}{dx}f(x) - \frac{d}{dx}g(x)\end{aligned}$$

🤔 All this says is: if the function has pieces that are added or subtracted you can take the derivative of each individual piece and add those derivatives.

The $\frac{d}{dx}$ notation can be a bit messy here, the prime notation is a bit better:

$$\begin{aligned}[f(x) + g(x)]' &= f'(x) + g'(x) \\ [f(x) - g(x)]' &= f'(x) - g'(x)\end{aligned}$$

Example

$$f(x) = x^3 - x^2 - 15$$

Rules for differentiation (3)

Constant Multiplier Rule



Rules for differentiation (3)

Constant Multiplier Rule

The derivative of a constant c multiplied by a function $f(x)$ is the same as the constant multiplied by the derivative:

$$\frac{d}{dx}[kf(x)] = k\frac{d}{dx}f(x)$$

 **Example**

Rules for differentiation (3)

Constant Multiplier Rule

The derivative of a constant c multiplied by a function $f(x)$ is the same as the constant multiplied by the derivative:

$$\frac{d}{dx}[kf(x)] = k \frac{d}{dx}f(x)$$

Example

$$y = x^2 + \sqrt{2}x - \frac{8}{x} + 4$$

Derivative of logs & exponents

Two special derivatives:

- $\frac{d}{dx}(e^x) = e^x$

- $\frac{d}{dx}\ln(x) = \frac{1}{x}$

Summary of differentiation rules

Differentiation Rule	$f'(x)$ notation	$\frac{d}{dx}$ notation
Constant Rule	$f(x) = c \Rightarrow f'(x) = 0$	$\frac{d}{dx}[c] = 0$
Power Rule	$f(x) = x^n \Rightarrow f'(x) = nx^{n-1}$	$\frac{d}{dx}[x^n] = nx^{n-1}$
Sum Rule	$(g(x) + h(x))' = g'(x) + h'(x)$	$\frac{d}{dx}[g(x) + h(x)] = \frac{d}{dx}g(x) + \frac{d}{dx}h(x)$
Difference Rule	$(g(x) - h(x))' = g'(x) - h'(x)$	$\frac{d}{dx}[g(x) - h(x)] = \frac{d}{dx}g(x) - \frac{d}{dx}h(x)$
Constant Multiplier Rule	$(c \cdot g(x))' = c g'(x)$	$\frac{d}{dx}[c g(x)] = c \frac{d}{dx}g(x)$

And two special derivatives: $\frac{d}{dx}(e^x) = e^x$ and $\frac{d}{dx}\ln(x) = \frac{1}{x}$.

There's more differentiation rules (product, quotient, chain rule).

They're fun, but we won't be reviewing them.

Higher order derivatives

- We can continue to take derivatives of derivatives.
- The second derivative becomes the rate of change of the rate of change.
- If we interpret the first derivative as velocity, then the second derivative is acceleration.
- Notation:
 - Function: $f(x) = y$
 - 1st derivative: $f'(x)$ or $\frac{dy}{dx}$
 - 2nd derivative: $f''(x)$ or $\frac{d^2y}{dx^2}$
 - nth derivative: $f^{(n)}(x)$ or $\frac{d^ny}{dx^n}$

Derivatives practice

Solve individually and then check with a partner.

1. Which rules should you use to take these derivatives?

$$\text{A) } f(x) = 3x^4 \qquad \text{B) } y = 4x^2 + 3x - 16$$

2. Find the derivatives of these functions

$$\text{A) } y = 3x^2 \qquad \text{B) } h(x) = 2x(x^2 + 3x) \qquad \text{C) } g(y) = 2\sqrt{y} - e^y + 20 \ln(y)$$

3. Find the 3rd derivative of:

$$G(z) = 3z^4 - 8z^3 + 2z - 19$$

Derivatives practice

 Let's see a solution!

2.B) Find the derivative

$$h(x) = 2x(x^2 + 3x)$$

Derivatives practice

 Let's see a solution!

2.C) Find the derivative

$$g(y) = 2\sqrt{y} - e^y + 20 \ln(y)$$

Derivatives practice

 Let's see a solution!

3. Find the 3rd derivative of $G(z) = 3z^4 - 8z^3 + 2z - 19$.

Derivatives solution

1. Which rules should you use to take these derivatives?

$$\text{A) } f(x) = 3x^4 \qquad \text{B) } y = 4x^2 + 3x - 16$$

- a. Power Rule (with the Constant Multiplier Rule)
- b. Sum/Difference Rule, Power Rule, Constant Multiplier Rule, and Constant Rule (for -16)

2. Find the derivatives of these functions

$$\text{A) } y = 3x^2 \qquad \text{B) } h(x) = 2x(x^2 + 3x)$$

$$\text{a. } y' = 3 \cdot 2x^{(2-1)} = 6x$$

$$\text{b. First expand: } h(x) = 2x \cdot x^2 + 2x \cdot 3x = 2x^3 + 6x^2$$

$$\text{Then differentiate term by term: } h'(x) = 2 \cdot 3x^{(3-1)} + 6 \cdot 2x^{(2-1)} = 6x^2 + 12x$$

Derivatives solution

2. Find the derivative of this function

$$g(y) = 2\sqrt{y} - e^y + 20 \ln(y)$$

Rewrite $\sqrt{y}$ as $y^{1/2}$:

$$\begin{aligned} g(y) &= 2y^{1/2} - e^y + 20 \ln(y) \\ g'(y) &= 2 \cdot \frac{1}{2} y^{(1/2-1)} - e^y + 20 \cdot \frac{1}{y} \\ &= \frac{1}{\sqrt{y}} - e^y + \frac{20}{y} \end{aligned}$$

Derivatives solution

3. Find the 3rd derivative of:

$$G(z) = 3z^4 - 8z^3 + 2z - 19$$

$$G'(z) = 3 \cdot 4z^{(4-1)} - 8 \cdot 3z^{(3-1)} + 2 \cdot 1z^{(1-1)} - 0 = 12z^3 - 24z^2 + 2$$

$$G''(z) = 12 \cdot 3z^{(3-1)} - 24 \cdot 2z^{(2-1)} + 0 = 36z^2 - 48z$$

$$G'''(z) = 36 \cdot 2z^{(2-1)} - 48 \cdot 1z^{(1-1)} = 72z - 48$$

Wrapping up...

Get some extra practice!

Problem set



What we covered today

-  Slope as a rate of change
-  Limits
-  Derivatives and rate of change
-  Tangent lines
-  Rules for differentiation
-  Constant rule
-  Power rule
-  Sum and difference rule